

Machine Learning-Based Intrusion Detection System for Encrypted Attacks

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ABSTRACT:

In today's society, information and communication technology is developing rapidly. With the gradual maturity and popularization of encryption technology, more and more malicious attacks are also using encryption technology to evade the scrutiny of traditional traffic detection systems. Therefore, accurate identification of encrypted attacks has become a research hotspot in the international community. This paper proposes an encrypted traffic detection method based on convolutional neural network (CNN) technology to address the issues of tedious steps and low recognition accuracy in manually extracting traffic features. This method does not require manual or expert feature extraction, and can automatically extract advanced features through CNNs, which are then fed into XGBoost classifiers for classification processing. On the basis of the above methods, this article designs and implements an encrypted traffic intrusion detection system, which is divided into five parts: traffic collection, data processing, model detection, data visualization, and traffic blocking. Reasonable explanations and technical introductions are provided for these modules.

Keywords: intrusion detection; encrypted traffic; CNN; network security.

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INTRODUCTION

The 1st century belongs to the information age, and in today's society, information and communication technology is developing rapidly [1-2]. Information technology used for human comfort, such as cell phone, computer, e-watch [3], smart-home [4-5], robotics [6], etc. According to relevant report statistics, by June 2023, the number of Internet users in China has reached 1011 million, an increase of 21.75 million over December 2022, and the Internet penetration rate has reached 71.6%, 1.2 percentage points higher than December 2022. At the same time, Internet traffic has also increased significantly. Figure 1 shows the change of Internet access traffic from 2020 to the first half of 2023. By the first half of 2023, Internet access traffic had reached 103.3 billion GB, with a year-on-year growth of 38.7%.



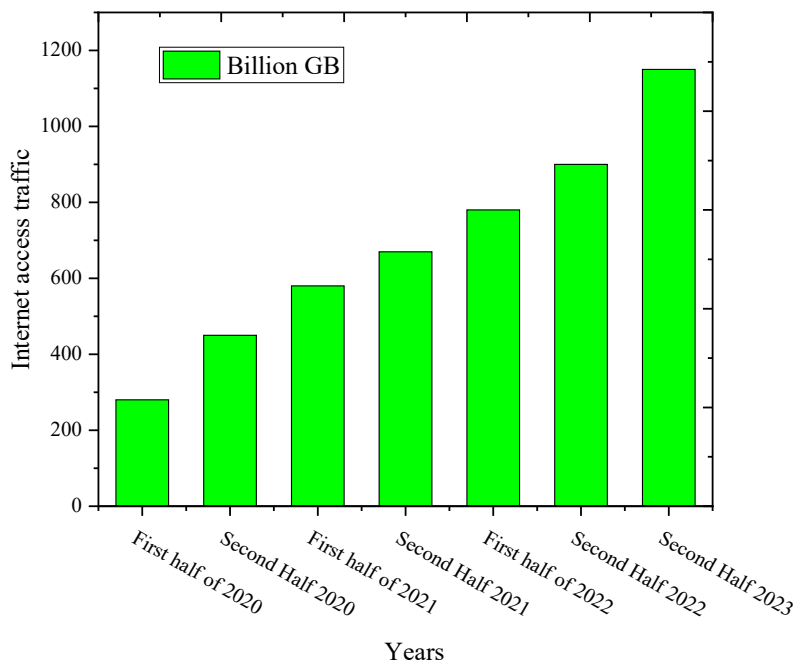


Figure 1. Changes in Internet Access Traffic

Individuals derive significant convenience from Internet services; however, they remain oblivious to the fact that their personal information and other security concerns are under constant attack. In AI era, a lot of information slots such digital twin technology, big data, cloud computing are useful for information communication [7-9]. These threats typically comprise two components: firstly, a significant number of applications excessively gather user information, and when this private data is disclosed or maliciously exploited, the repercussions can be extremely severe; The second component involves illegal entities unlawfully collecting and trading personal information. These occurrences have caused a chain of network information security problems, leading to an increasing focus on information protection in society. As a result, encrypted traffic has become prevalent. Encrypted traffic is the data that is produced by a particular encryption technique, typically used to encrypt only the plain text content during network transmission. Obtaining plaintext and encryption keys only from the encrypted ciphertext material is a challenging task. Encrypted traffic can efficiently safeguard user privacy data and prevent password analysis attacks.

According to a report by Gardner Corporation [10], over 80% of the traffic in 2019 was encrypted, with HTTPS (Hyper Text Transfer Protocol over SecureSocket Layer) traffic accounting for the majority. In 2019 alone, over one million HTTPS encryption certificates were issued daily. Encryption technology is a double-edged sword. On the one hand, it protects the privacy of user data and avoids the theft of information by a third party. On the other hand, it also contributes to the growth of malicious traffic. Malicious traffic also uses encryption technology in order to avoid traffic detection, especially some network attacks and illegal information dissemination. Because of the use of encryption technology, it acts recklessly. In order to better ensure Internet security and quality of service, it needs to classify encrypted traffic or intrusion detection. The previous detection methods for unencrypted traffic are no longer applicable to encrypted traffic [11]. For example, traditional packet detection methods only evaluate packet header information such as source IP address, destination IP address, and port number.

The deep packet detection method examines more comprehensive data and metadata related to a single data packet. This method requires analyzing the specific content of traffic packets and using feature matching techniques to match the features of the packets with existing feature rule libraries, thereby achieving flow detection. However, this method violates user privacy by parsing packet content, and the content characteristics of encrypted traffic will undergo significant changes. There is also a port number-based detection method [12], which assumes that the majority of network traffic uses the port number assigned by ICANN (The Internet Corporation for Assigned Names and



Numbers), and matches the corresponding traffic based on the port number. However, the subsequent occurrence of port camouflage, port obfuscation, and tunneling techniques quickly rendered this method ineffective.

Deep packet detection methods are no longer effective, and people are turning their attention to machine learning (ML) because encryption technology only encrypts packet information and does not involve encryption of flow statistical features. Therefore, ML methods based on flow statistical features are less affected by encrypted traffic. Deep learning methods in ML can also extract deeper and more effective encrypted traffic features without the need for manual intervention. There are currently several issues in the field of intrusion detection, including:

- The significant surge in encrypted data has complicated the network environment, making standard traffic detection methods unsuitable. Originally, they analyzed plaintext communications without encryption.
- Some approaches for identifying encrypted communication can only recognize specific types of traffic, which is inefficient; others need manual extraction of traffic characteristics, which is laborious.
- Generally, more complicated trained models perform better but require more storage and resources. This hinders model deployment on different hardware platforms. Thus, intrusion detection requires a model that balances complexity and performance on diverse hardware platforms.

LITERATURE REVIEW

The identification of encrypted traffic has gradually attracted people's attention and become an important research direction in the field of network security. After recent years of research, the current methods for identifying encrypted traffic in networks are mainly divided into the following types: supervised machine learning methods, semi supervised machine learning methods, and deep learning-based methods.

Supervised ML Methods

Sun et al. [13] proposed a hybrid method for classifying encrypted traffic. Firstly, use signature matching method to identify the SSL VTLS protocol, and then use naive Bayesian machine learning algorithm to identify the encrypted application protocol. The experimental results showed that this method can accurately identify the vast majority of SSL/TLS traffic. Okada et al. [14] analyzed the changes in flow characteristics caused by traffic encryption and created their own dataset using point-to-point tunneling protocol (PPTP) and IPsec tunnel encryption for HTTP, FTP (File Transfer Protocol), SSH (Secure Shell), and SMTP application protocol (Simple Mail Transfer Protocol). Therefore, a standard classifier can be used to classify the transformed traffic, and the author used a modified Naive Bayes algorithm to validate their method. The experimental results show that compared with existing methods, the author's method has improved the recognition accuracy of encrypted traffic by 28.5%. In addition, by eliminating features that do not conform to Gaussian distributions, using the optimal combination of flow features for recognition can achieve high-precision recognition with less computational overhead.

Arndt et al. [15] focused on the classification of encrypted traffic, without using IP address, port number, and payload information, but compared it with C4.5, k-means, and multi-objective genetic algorithm (MOGA). They used the classification of SSH protocol as an example in their research, focusing on the accuracy and robustness of the algorithm, and used three different public and private datasets to verify the stability of the algorithm. The experimental results show that the C4.5 method provides the fastest and most readable model, while MOGA greatly reduces the complexity of the k-means clustering algorithm. The C4.5 algorithm has the best robustness, and the false alarm rate of the MOGA algorithm is very low. In addition, studies have shown that continuous sampling of training data is not better than random sampling, but the impact of training data on the performance of classifiers in identifying traffic captured by different networks is crucial.

Alshammari et al. [16] have published multiple papers on the use of various supervised ML methods for traffic classification. They focus on identifying the traffic of SSH, Skype, and Gtalk, using traffic



characteristics without port numbers, IP addresses, or loads. The dataset used comes from both public and private datasets. The category results of public datasets are obtained through port numbers, while private datasets include payloads and are labeled using traffic classification tools. The experimental results show that among many supervised machine learning methods, the C4.5 machine learning method has the best performance.

Semi Supervised Learning Algorithms

Bemaille et al. [17] uses traffic clustering to detect SSL (Secure Sockets Layer) encryption applications, which consists of three steps: first, they use a clustering algorithm (Gaussian mixture model) to detect SSL connections, which detects the size of packets and the direction of the initial packets connected. After the SSL traffic is identified, the first packet of the connection is identified; In the third step, the clustering algorithm uses the size of packets to detect underlying applications. The datasets used include HTTP, POP3 (Post Office Protocol Version 3), FTP, BitTorrent, and the eDonkey application protocol encrypted with SSL. The author also suggests that using a combination of clusters and port numbers to distinguish applications within a cluster can provide better results than using a cluster alone.

Backquet et al. [18] used MOGA to select flow feature subspaces and parameters as clustering algorithms to detect encrypted traffic, and then adopted a layered k-means algorithm to improve recognition accuracy. The author evaluated the performance of the algorithm on a private dataset on a university campus, using SSH as a representative of encrypted traffic to obtain the final result from the captured packet load. Based on previous research, the author believes that the selection of features and the number of clusters greatly affect the overall accuracy. Therefore, the author selected four objectives for the genetic algorithm: cluster cohesion, cluster separation, number of clusters, and usage of flow features. The results showed that only 14 out of 38 flow features were utilized, and the use of hierarchical k-means algorithm can improve recognition performance.

Zhang et al. [19] used harmonic averaging to lessen the influence of random starting cluster centres to increase the k-means clustering algorithm's performance in categorising encrypted traffic. The author tested their method on two datasets: one with solely SSH and non-SSH data, and the other including Skype, QQ, SSH, SSL, and MSN traffic. This classification algorithm performs better than earlier methods, according to experiments.

Deep Learning-Based Methods

Wang et al. [20] proposed a method of using CNNs to solve the problem of encrypted traffic recognition, which includes feature extraction, feature selection, and traffic classifier, together forming an end-to-end framework. The author claims that this is the first application of end-to-end methods in the field of encrypted stream classification. This method has been validated on the public dataset ISCXVPN non-VPN. Among the 12 evaluation indicators in the experimental results, 11 are superior to the current method, indicating the effectiveness of the author's proposed method.

He et al. [21] suggested an image-based technique for precise encrypted network traffic classification. This method converts the first several non-zero payload sizes of a session into grey and white images and classifies them using a two-dimensional CNN. It automatically extracts, selects, and classifies encrypted network traffic with a lightweight feature. His technique was validated using the public dataset ISCXVPN2016. This method's F1 score on traditional encrypted traffic classification is 97.73% and on VPN classification is 99.55%, according to experiments.

Wang et al. [22] proposed a framework for implementing CNN and Stacked Autoencoders (SAE). The framework uses CNN to extract advanced features from raw network traffic and encodes 26 statistical features directly calculated from raw traffic using SAE. These statistical features can be used to supplement the information loss caused by pruning. Afterwards, the output from the encoder in CNN and SAE is combined into new advanced features, including information from pruned raw traffic and statistical functions. Finally, these new advanced features are used to classify encrypted traffic. The authors validated the feasibility of the framework using the ISCXVPN2016 traffic dataset, and its F1 value reached 0.98. It can improve the performance of encrypted traffic classification. In addition, new high-level features generated by combining features extracted from CNNs and stacked autoencoders can effectively represent traffic of different categories. More importantly, the author



believes that this work is unique in the field of encrypted traffic classification, as it is the first time using raw traffic and statistical features as inputs to the model.

Based on the above analysis, after years of research and development in academia and industry, encrypted network traffic classification has achieved a lot of results. However, ML based traffic classification often requires manual selection of traffic features, and feature extraction requires expert experience. The rationality of feature extraction will directly affect the final classification effect. Combining deep learning with encrypted network traffic classification to automatically extract features from the original traffic and classify them has gradually become a popular direction. However, there are still some problems:

- How to handle the original traffic appropriately and correctly. The information carried by the raw traffic collected from the real environment is irrelevant to the recognition task and may even affect the final recognition result. Therefore, it is necessary to correctly screen out these irrelevant information.
- The traffic classifier model requires too much platform resources. The current model trained through neural networks has a high complexity, occupies huge computing resources, and has long training and inference times. With the continuous expansion of the Internet boundary, a variety of Internet terminals that meet the needs continue to appear. The hardware conditions of these terminals are limited. Therefore, how to reduce the computing resource requirements of the model has become an urgent problem to be solved.

METHODOLOGY

This study introduces the specific method of encrypted traffic intrusion detection based on pruning CNNs, which is divided into two parts: data preprocessing and encrypted traffic recognition. In the data preprocessing section, it is necessary to first remove information that may cause overfitting of the model or be irrelevant to the recognition task, and then convert the original traffic into a format that CNNs can accept. In the encrypted traffic recognition section, this article uses pruning techniques to prune the constructed CNN model, retaining only those parameters that have a high degree of impact on the model [23].

Encrypted Stream Intrusion Detection

As shown in Figure 2, in order to better manage and analyze encrypted traffic, we used traffic mirroring technology. Switched Port Analyzer can generally be used for network monitoring and analysis. Traffic mirroring is mainly used to detect attacks and can also achieve partial defense against attacks. Traffic mirroring can detect the full traffic data of L4. Protocol layer attacks can be detected, such as TCP SYN flooding, IP source address spoofing, ICMP routing redirection attacks, TCP session hijacking attacks, and application layer attacks, such as SQL injection and HTTP GET attacks. The mirrored traffic is output to the PCNN (pruning convolutional network) model proposed in this article for encrypted traffic identification. This technology is low-cost, does not require any additional network equipment, has no impact on switch performance, and can ensure the normal operation of existing services. The focus of this framework is the PCNN method proposed in this article, which preprocesses the original traffic data, removes irrelevant information, converts it into a format suitable for CNN, constructs a CNN + XGBoost model, prunes model parameters, and finally uses the pruned model to identify and classify encrypted traffic.

Data Set

There are not many publicly available datasets for encrypted traffic detection. How many datasets have we found about encrypted traffic detection through consulting public literature. For example, Lashkari et al. [24] published a dataset on Tor traffic, which contains 8 different labels corresponding to the captured 8 different types of traffic: web traffic, audio traffic, chat, video traffic, email, VoIP, P2P, and file transfer. Bissias et al. provided a dataset on HTTP encryption, which includes malicious attacks hidden in HTTPS encrypted traffic and normal traffic [25].



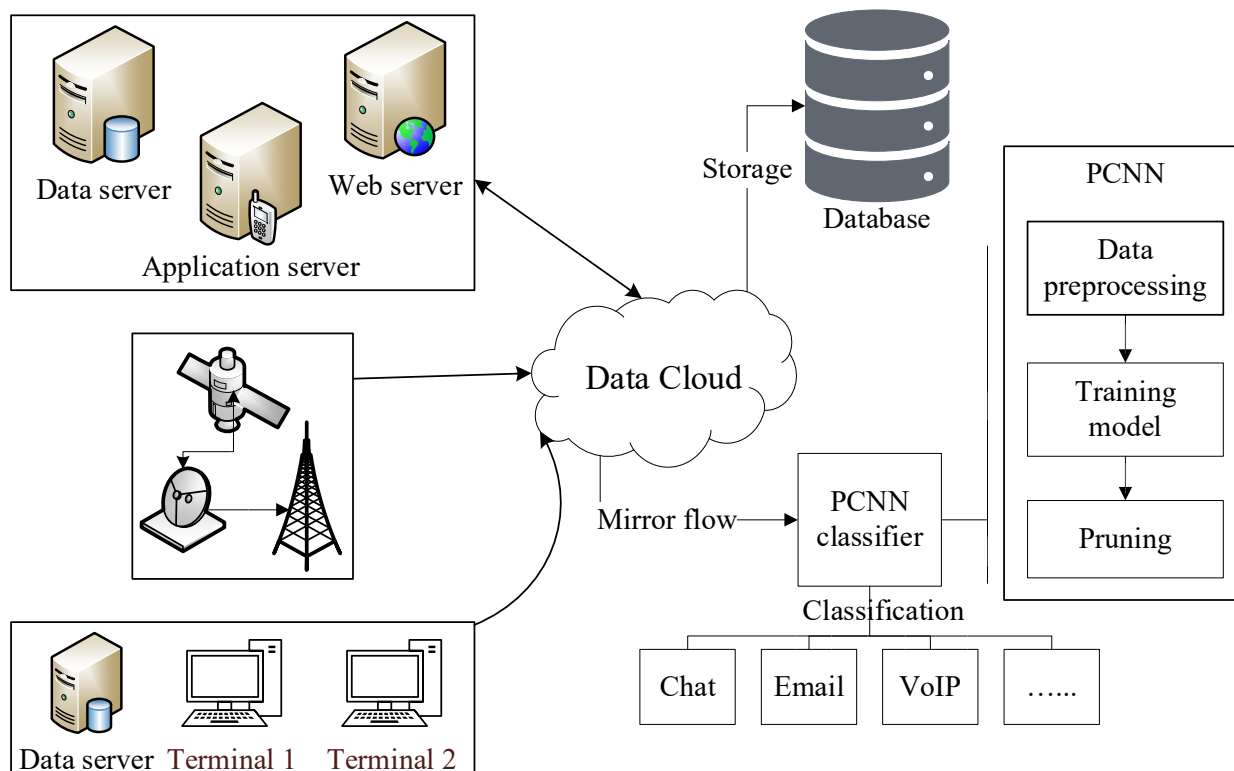


Figure 2. Encryption Traffic Intrusion Detection Framework

Data Preprocessing

Because the selected dataset comes from a real environment, it definitely contains some redundant information that is useless for traffic recognition and may even affect the inference of the model. In order to convert the original encrypted traffic into an input format suitable for CNNs, preprocessing operations need to be performed on the original traffic. This article refers to the format of the MNIST dataset, which converts the original traffic into an idx file format, with each image size of 28x28. The data preprocessing process is shown in Figure 3, and the key steps include:

Traffic split: It divides continuous raw traffic into discrete sessions. Currently, the common ways of representing traffic are sessions and streams, where streams refer to all traffic packets with the same quintuple, that is, source IP, source port, destination IP, destination port, and transport layer protocol. The bidirectional flow forms a conversation, where the source IP, source port, destination IP, and destination port in a five tuple can be interchanged. Some studies have pointed out that using a session-based approach for traffic segmentation is more effective than using a flow-based approach. Therefore, this article uses a session-based approach with the help of SplitCap open-source tools to segment raw traffic. SplitCap can divide a large file into several small files, with each session corresponding to a small file. Its characteristics are fast speed, convenient use, and customizable operations according to different standards during segmentation

Remove irrelevant packets: Taking TCP connection as an example, there are 6 control bits in a TCP message, namely URG, ACK, PSH, RST, SYN, and FIN. When the ACK, SYN, or FIN flags in the message are 1, they are used for TCP three handshake or four wave, and there is no valid information that needs to be removed. It is also necessary to check whether the data part of the TCP packet is empty, and if it is empty, it needs to be removed. When the message belongs to the DNS protocol, it also needs to be removed, which is independent of the traffic identification task.

Remove Ethernet header: The Ethernet header contains the MAC address, which is used to identify specific network nodes, but is irrelevant in traffic identification.

Mask IP address: Because the dataset is captured using a limited number of hosts and servers, the trained model may treat IP addresses as traffic characteristics, which may lead to overfitting or false



positives. So, it is necessary to set both the source IP address and the destination IP address to 0.0.0.0. This ensures that the model does not mistake IP addresses for traffic characteristics.

Determine packet length: The CNN model requires that the size of the input data be fixed, so a truncation length needs to be set. When the packet length is greater than the truncation length, the packet is truncated to that length; When the packet length is less than the truncation length, add 0x00 at the end of the packet. This ensures that the packet length in all sessions is consistent.

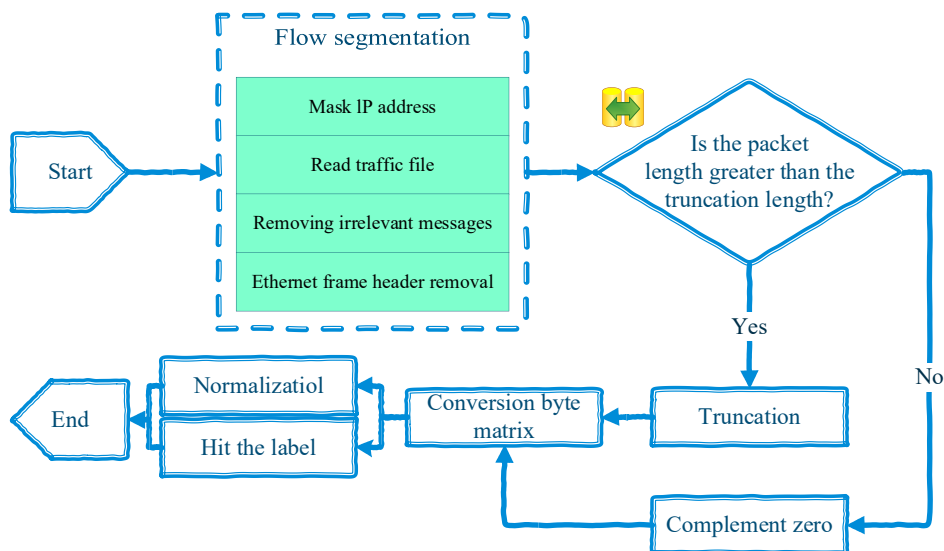


Figure 3. Data Preprocessing Process

In order to visually distinguish the differences between various encrypted traffic, this article converts the processed traffic files into grayscale images in binary form, such as 0x00 representing white and 0xff representing black. The graph shows the visualization results of 12 types of traffic, with one randomly selected for each type.

Evaluation Indicators

In classification tasks, the calculation of each indicator is based on the classification results of positive and negative samples. Usually, there are three evaluation indicators to measure the effectiveness of the trained model, namely precision, recall, and F1 value. There is also an evaluation method called confusion matrix, also known as error matrix. The confusion matrix is generally composed of n rows and n columns in the form of a matrix. For example, in binary classification problems, the corresponding confusion matrix is a 2x2 matrix, as shown in Table 1.

Table 1. Binary Confusion Matrix

Binary confusion matrix		Predicted	
		Positive	Negative
Actual	Positive	TP	FP
	Negative	FN	TN

Table 1 defines TP as the projected positive, which coincides with the actual positive. FP stands for false positive, which means that it predicts a positive outcome when in reality it is negative. FN stands for false negative, which occurs when a prediction indicates a negative outcome, while the actual outcome is positive. TN stands for true negative, indicating that the expected outcome is negative and matches the actual outcome. The calculation technique for accuracy, recall, and F1 value can be derived from the provided confusion matrix:

Precision (P_r): The proportion of correctly identified traffic types to the entire test dataset, is:



$$P_r = \frac{TP}{TP+FP} \quad (1)$$

Recall rate (R_c): The proportion of correctly predicted traffic types to all such types in the entire test dataset, also known as recall rate, is:

$$R_c = \frac{TP}{TP+FN} \quad (2)$$

$F1$ -value: derived from precision and recall, it is a comprehensive evaluation index for the overall evaluation of precision and recall, as well as the accuracy of the model, is as:

$$F1 = \frac{2 \times P_r \times R_c}{P_r + R_c} \quad (3)$$

EXPERIMENTAL RESULTS

The ratio of the weights that have been pruned to the weights of all of the neurons in the CNN model is what is meant by the term "pruning rate," as depicted in Figure 4. This study investigated two different approaches of pruning. In this method, weights that have larger absolute values are eliminated depending on a predetermined pruning rate. In the second method, weights that have lower absolute values are eliminated depending on a predetermined pruning rate. It is clear from looking at Figure 4 that the green line is decreasing at a faster rate as the pruning rate continues to climb. Method 1 was selected for this article because it demonstrates that it has superior performance and that weights with greater absolute values contribute more to the results that are produced by the model.

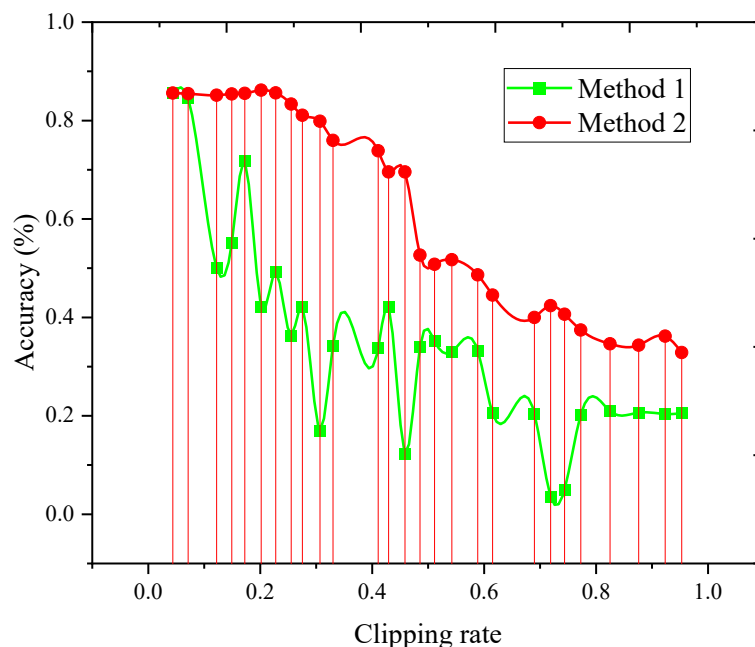


Figure 4. Comparison of the Two Pruning Methods

In order to verify the effectiveness of pruning, this article compared the effects before and after pruning, and selected a pruning ratio of 0.45, as shown in Table 2. It can be seen that the recall rate of Chat type increased from 0.14 to 0.95, and the $F1$ value increased from 0.25 to 0.97. The



evaluation metrics for both encrypted and non-encrypted traffic have been improved, indicating that the pruned model can better learn traffic characteristics and enhance recognition ability. At the same time, during the model storage process, zero weights are not stored, reducing the storage size from 48.25MB before pruning to 32.19MB, reducing storage resource consumption by about 33%, and lowering the storage resource requirements of the model. The detection time on the test dataset was reduced from 360ms before pruning to 205ms, and the detection time consumption was reduced by about 50%, reducing some algorithm overhead.

Table 2. Comparison Before and After Clipping

Service type	Before clipping			After clipping		
	P_r	R_c	$F1$	P_r	R_c	$F1$
Chat	1.0	0.14	0.25	1.0	0.95	0.97
Email	0.94	0.17	0.29	0.99	0.88	0.93
File transfer	0.31	1.0	0.47	0.75	1.0	0.86
Streaming	0.99	0.72	0.83	0.96	0.87	0.91
Torrent	0.97	0.99	0.98	0.98	0.99	0.98
Voip	0.63	0.62	0.62	1.0	0.81	0.90
VPN-Chat	0.92	0.91	0.91	0.85	0.91	0.88
VPN-Email	0.75	0.85	0.8	0.89	0.95	0.92
VPN-File transfer	0.97	1.0	0.98	0.97	1.0	0.98
VPN-Streaming	0.99	1.0	0.99	0.98	1.0	0.99

On the basis of the above work, this paper conducted binary classification experiments on the dataset to verify the effectiveness of the proposed algorithm in identifying encrypted malicious traffic. As there are only two types of traffic in the dataset, namely black and white samples, the XGBoost algorithm needs to output binary classification. The experimental results are shown in Table 2.

Comparison of Related Work

In order to better explore the effectiveness of the method proposed in this article, several other classic methods for encrypted traffic recognition were selected for comparison, as shown in Table 3. It can be seen that the algorithm proposed in this article has significant improvements in accuracy, recall, and F1 value compared to other algorithms. This article speculates that pruning techniques remove invalid features that lead to model inference errors, making the model more focused on features with greater impact and increasing the recognition accuracy of the model.

Table 3. Comparison of Proposed Method with Related Work

Methods	Accuracy (%)	Recall rate	F1-value
ID CNN [20]	0.89	0.89	0.89
2D-CNN [21]	0.91	0.91	0.91
SAE [26]	0.92	0.92	0.92
CNN+LSTM [27]	0.91	0.91	0.91
This work	Before clipping	0.90	0.88
	After clipping	0.94	0.93

CONCLUSION

This study proposes an encrypted traffic recognition method based on pruning CNN technology. This method does not require manual or expert feature extraction, and can automatically extract advanced features through CNNs. The method is divided into two parts: data preprocessing and encrypted traffic recognition. In the data preprocessing section, it is necessary to first remove information that may cause overfitting or be irrelevant to the recognition task, such as IP addresses, and then convert the original traffic into a format that CNNs can receive. In the encrypted traffic recognition section, this article uses pruning techniques to prune the constructed CNN model, retaining only those parameters that have a high degree of impact on the model. The features



extracted by the CNN model will be passed into the XGBoost classifier to complete the final classification task. The final experimental section proves that weights with larger absolute values have a greater impact on the model than weights with smaller absolute values. Therefore, set the weight with a smaller absolute value to 0 (the optimal clipping ratio is 0.45). The experimental results conducted on the public dataset validated the reliability of the method: accuracy reached 94%, improved by 5%, recall reached 93%, improved by 4%, F1 value reached 93%, improved by 4%. Some datasets have significant differences in traffic distribution, which can make traffic recognition algorithms more inclined towards traffic types with a higher proportion. Unbalanced types can directly affect classification accuracy. Therefore, how to solve the problem of data imbalance is an important direction in the field of human intrusion detection in the future.

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